

Beijing University of Aeronautics and Astronautics



Regional Centre for Space Science and Technology Education in Asia and the Pacific

CFD Analysis of NACA 2412 Airfoil

Mesh Development, Solver Selection, and Stall Prediction

Aerodynamics

Prof. Chen Song 陈松

By

Marwane OUFRID 马万
LS2525201

Venue: Hangzhou, China

Date: June 2026

Abstract

This study presents an open-source computational fluid dynamics workflow for the NACA 2412 airfoil using Gmsh, SU2, and ParaView. A single formal grid-convergence family with 152,354, 288,996, and 550,284 elements was constructed at a chord Reynolds number of 3.1×10^6. At 0°, drag converged monotonically with observed order $p = 3.60$; the 288,996-element middle grid differed from the Richardson-extrapolated drag by 1.13%, while lift varied by only 0.0196% across the three meshes. The corresponding fine-pair and coarse-pair Grid Convergence Indices were 0.443% and 1.401%, and the selected production grid required 45.9% of the fine-grid solver work. Aerodynamic calculations used incompressible Reynolds-averaged Navier–Stokes equations with the SST-2003m turbulence model and Langtry–Menter transition correlation for the attached and near-stall branch from -5° to 15°. A spanwise-periodic SA-DDES pilot was retained as the best tested representation of the post-stall trend from 16° to 20°. The assembled polar gives $C_L = 0.2289$ and $C_D = 0.00523$ at 0°, maximum $C_L/C_D = 89.68$ at 4°, and maximum lift of approximately 1.62 at 15°, followed by a sharp post-stall reduction. ParaView fields at 0°, 4°, 15°, and 20° compare pressure, velocity, streamline, and wake behavior. The resulting archive keeps the geometry, meshes, solver settings, histories, post-processing figures, and publication sources together, demonstrating the value of open tools for auditable aerodynamic research.

Keywords: NACA 2412; CFD; Gmsh; SU2; ParaView.

摘要

本研究采用 Gmsh、SU2 和 ParaView 建立了 NACA 2412 翼型的开源计算流体力学工作流程。在弦长雷诺数 3.1×10^6 条件下，构建了包含 152,354、288,996 和 550,284 个单元的统一网格收敛族。在 0° 攻角下，阻力呈单调收敛，观测阶数为 $p = 3.60$；288,996 单元的中等网格与 Richardson 外推阻力的差异为 1.13%，三套网格之间的升力总变化仅为 0.0196%。细网格对和粗网格对的网格收敛指数分别为 0.443% 和 1.401%，所选生产网格的求解工作量为细网格的 45.9%。附着流和近失速分支采用不可压缩 Reynolds 平均 Navier–Stokes 方程、SST-2003m 湍流模型以及 Langtry–Menter 转捩相关模型，覆盖 -5° 至 15°。展向周期 SA-DDES 试算被保留为 16° 至 20° 失速后趋势的最佳测试表示。组合极曲线在 0° 给出 $C_L = 0.2289$、$C_D = 0.00523$，在 4° 得到最大升阻比 89.68，并在 15° 达到约 1.62 的最大升力，随后出现明显的失速后升力下降。ParaView 后处理比较了 0°、4°、15° 和 20° 下的压力、速度、流线和尾迹结构。最终归档同时保留几何、网格、求解器设置、收敛历史、后处理图像和论文源文件，体现了开源工具在可审查气动研究中的价值。

关键词： NACA 2412；计算流体力学；Gmsh；SU2；ParaView。

Contents

1 Introduction **1**

2 Geometry and Computational Method **1**

2.1 Airfoil and flow conditions 1

2.2 Open-source workflow 1

3 Gmsh Mesh Development **2**

3.1 Near-wall strategy 2

3.2 Discarded trailing-edge fan concept 2

3.3 Systematic three-grid family 4

3.4 Formal grid-convergence verification 5

3.5 Accuracy–cost benchmark 7

4 SU2 Solver Selection and Convergence **8**

4.1 Iterative model-selection process 8

4.2 Post-stall model screening 9

4.3 Residual histories 9

5 Aerodynamic Results **10**

5.1 Construction of the reported polar 10

5.2 Comparison with thin-airfoil theory 11

5.3 Overlay on the historical NACA chart 12

6 ParaView Flow-Field Analysis **12**

7 Discussion of Stall Behavior **15**

8 Conclusions **15**

List of Figures

1 The earlier fan captured the geometric singularity, but it was not an efficient topology for a long aerodynamic wake. . . . 3
2 Production mesh at full-domain and airfoil scales. . . . 3
3 Detailed production-grid near-wall topology. . . . 4
4 Element composition of the systematic three-grid family. . . . 5
5 Near-airfoil topology across the systematic mesh family. . . . 5
6 Formal grid-convergence behavior for drag and lift at 0°. The production mesh is the middle grid. . . . 6
7 Coefficient sensitivity and normalized solver work for the systematic 0° mesh family. . . . 8
8 Pressure-residual histories from all 26 angle cases. . . . 9
9 Aerodynamic coefficients versus angle of attack. . . . 10
10 Polar representation and C_L/C_D versus incidence. . . . 11
11 CFD lift curve overlaid with thin-airfoil theory. . . . 11
12 Current CFD curves superimposed on the historical NACA 2412 chart [1]. . . . 12
13 ParaView fields at 0°: reference incidence. . . . 13
14 ParaView fields at 4°: maximum-C_L/C_D condition. . . . 13
15 ParaView fields at 15°: maximum-lift condition. . . . 14
16 ParaView fields at 20°: post-stall endpoint. . . . 14

List of Tables

1 Systematic grid-family definition and calculation size. . . . 4
2 Formal 0° grid-convergence results. . . . 6
3 Mesh accuracy–cost benchmark at 0° . . . 7
4 Aerodynamic coefficients for all 26 reported angles of attack. . . . 10

1. Introduction

Computational fluid dynamics (CFD) provides numerical approximations to the conservation equations governing fluid motion. For external aerodynamics, the principal outputs are surface pressure, viscous shear, lift, drag, and the development of separated wakes. These quantities are sensitive to geometry, domain extent, mesh resolution, boundary conditions, turbulence and transition models, numerical convergence, and the averaging interval used for unsteady flow. A credible airfoil study must therefore document the complete chain from coordinates to final aerodynamic coefficients rather than treating the solver as a black box [2, 3].

The NACA four-digit family remains useful for such studies because its geometry is compactly defined and extensive historical measurements exist. In NACA 2412, the first digit denotes a maximum camber of 2% chord, the second places that maximum at 40% chord, and the final two specify a maximum thickness of 12% chord. The profile provides positive lift at zero geometric incidence and a negative zero-lift angle, making it a useful test of both linear lifting theory and viscous stall behavior [1, 2].

This project also examines the research benefit of open software. Source-visible tools make numerical settings auditable, allow scripts and configuration files to be archived with results, and reduce licensing barriers to replication. Open programs do not automatically guarantee reproducibility, but they make it possible to inspect and rerun the full workflow when accompanied by organized data, versioned meshes, and explicit convergence evidence [4]. Gmsh, SU2, and ParaView form the open toolchain used here.

2. Geometry and Computational Method

2.1 Airfoil and flow conditions

The NACA 2412 coordinates were imported as an ordered point set and connected by a spline. The trailing-edge duplicate in the source coordinate list was removed for meshing so that the closed airfoil boundary remained topologically valid. The chord was normalized to $c = 1$.

The reference condition was a low-speed wind-tunnel comparison at $\mathrm{Re}_c = 3.1 \times 10^6$. The computational domain extended $10c$ upstream of the leading edge, $20c$ downstream of the trailing edge, and $10c$ above and below the chord line. The angle sweep contained 26 cases from -5° to 20°, inclusive. The no-slip airfoil wall and distant farfield boundary were retained across the production sweep.

2.2 Open-source workflow

Gmsh is a scriptable finite-element mesh generator with geometry, meshing, and post-processing capabilities [5]. It was used to preserve the airfoil spline, construct the farfield, create the structured near-wall block, and grade the unstructured triangular region. SU2 is an open-source multiphysics suite designed for analysis and aerodynamic design [6]. It was used for steady RANS, URANS, and high-angle model screening. ParaView provides a pipeline-based environment for large scientific datasets and was used for all field visualization [7].

The workflow was:

1. import and verify the NACA coordinates;
2. construct and quality-check the Gmsh domain and physical boundaries;
3. convert the mesh to SU2 format;
4. test solver formulations and turbulence/transition settings at $0°$;
5. run the angle sweep with one documented folder per case;
6. continue difficult cases and evaluate saved residual and coefficient histories;
7. post-process representative fields in ParaView;
8. generate plots and publication files from archived CSV and PNG outputs.

3. Gmsh Mesh Development

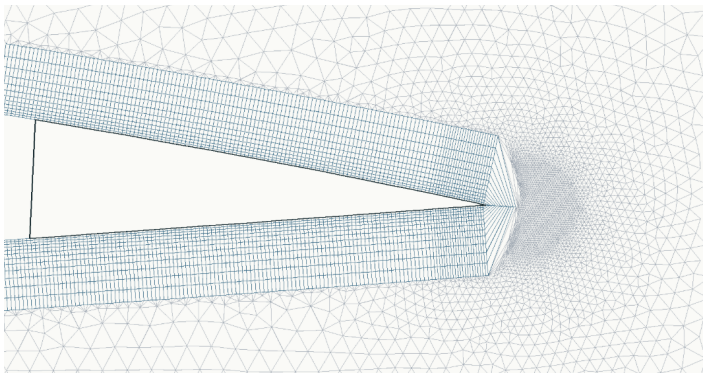
3.1 Near-wall strategy

The airfoil was surrounded by quadrilateral cells whose tangential lines follow the surface and whose transverse lines are approximately surface-normal. Thirty layers were used so that cell height increased smoothly away from the wall. The production first-layer height was $8.45 \times 10^{-6}c$, estimated for $y^+ \approx 1$ at $\mathrm{Re}_c = 3.1 \times 10^6$. The total boundary-layer block thickness was $0.020c$, with geometric growth ratio 1.23457. Strong local refinement was retained at the leading edge, while the trailing edge used a curved fan-like transition into the wake region.

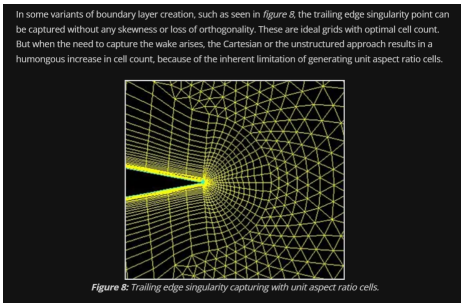
The complete farfield was filled with triangular elements. A circular refined zone, $5c$ in diameter and centered at $(x/c, y/c) = (1.5, 0)$, concentrated resolution around the airfoil and immediate wake. Element size then increased progressively toward the upstream boundary and distant corners.

3.2 Discarded trailing-edge fan concept

An earlier development mesh attempted to wrap the upper and lower boundary-layer blocks into a compact radial fan immediately behind the sharp trailing edge. This construction prevented surface-normal grid lines from crossing the airfoil and could maintain acceptable orthogonality close to the singularity. It was nevertheless rejected as the preferred wake topology.



(a) Discarded project candidate: compact radial fan followed by a dense isotropic triangular region.

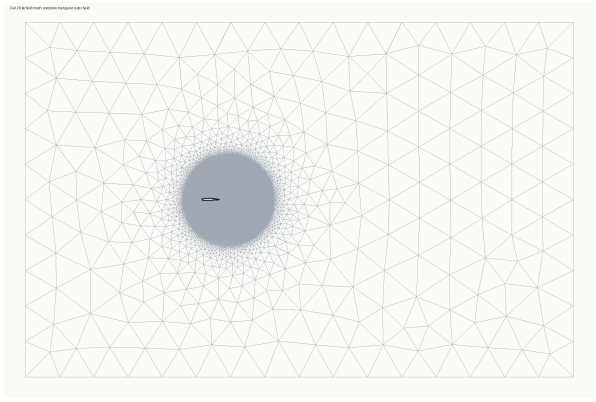


(b) Trailing-edge singularity example discussed by GridPro [8].

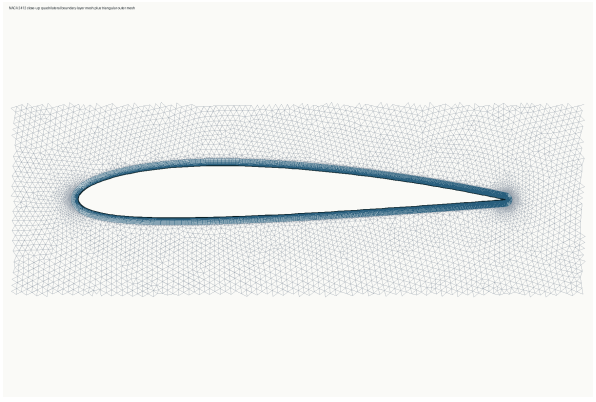
Figure 1: The earlier fan captured the geometric singularity, but it was not an efficient topology for a long aerodynamic wake.

The limitation is not simply visual. Radial spokes and near-unit-aspect-ratio triangles consume many cells around the trailing edge without aligning them with the downstream shear layers. The rapid transition from high-aspect-ratio boundary-layer quadrilaterals to the fan and then to isotropic triangles also creates avoidable size and orientation changes. GridPro notes that an unstructured or Cartesian treatment can resolve a trailing-edge singularity cleanly, yet extending the same unit-aspect-ratio resolution through the wake causes a very large increase in cell count [8]. The same source recommends wake-oriented loop or nesting concepts, in which stretched cells follow the shear layer and resolution can decay without propagating boundary-layer fineness across the complete downstream domain.

The retained project mesh therefore used only a compact trailing-edge transition together with a graded downstream refinement field. This was an affordable compromise for the 26-case campaign, but a future production mesh should replace the remaining radial character with a wake-aligned C-grid or loop topology, especially for high-angle separated flow.

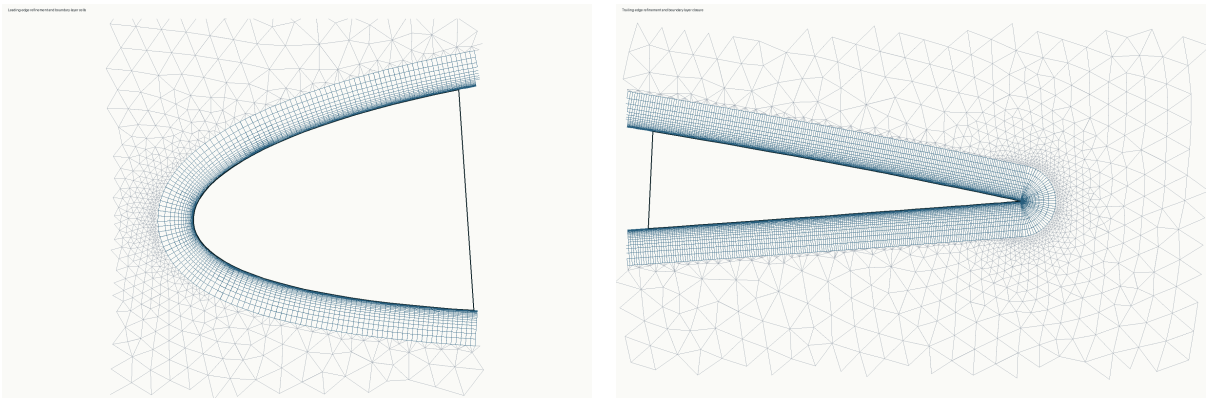


(a) Full production domain.



(b) Airfoil and concentrated near field.

Figure 2: Production mesh at full-domain and airfoil scales.



(a) Continuous leading-edge quadrilateral block. (b) Trailing-edge fan and wake transition.

Figure 3: Detailed production-grid near-wall topology.

3.3 Systematic three-grid family

One mesh family is used consistently for mesh description, formal grid-convergence verification, and the accuracy–cost assessment. The coarse, production, and fine grids contain 152,354, 288,996, and 550,284 elements, respectively. All three preserve the same physical domain, near-field and wake zones, 30-layer wall-normal construction, trailing-edge treatment, and Gmsh algorithm. Refinement is applied systematically through the target cell sizes, so the family is suitable for Richardson extrapolation and GCI evaluation.

Relative to the production grid, every target cell size was multiplied by $\sqrt{2}$ to create the coarse grid and divided by $\sqrt{2}$ to create the fine grid. This applies the same fundamental refinement rule throughout the near-wall, wake, and outer-flow regions rather than changing the topology between grids. The corresponding first-cell heights give approximately $y^+ = 1.41$, 1.00, and 0.71 on the coarse, production, and fine grids, respectively. All three values remain within the viscous-sublayer-resolved range.

Table 1: Systematic grid-family definition and calculation size.

Grid	Elements	Triangles	Quads	h/h_1	y_1/c	Iterations
Coarse	152,354	123,484	28,870	1.9005	1.1950×10^{-5}	5,500
Production	288,996	246,779	42,217	1.3799	8.4500×10^{-6}	5,467
Fine	550,284	486,474	63,810	1.0000	5.9751×10^{-6}	6,258

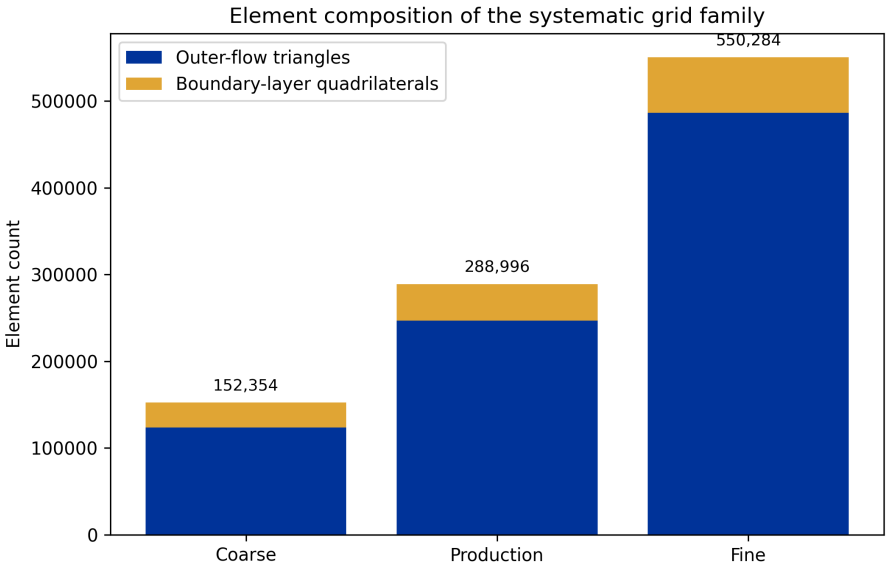
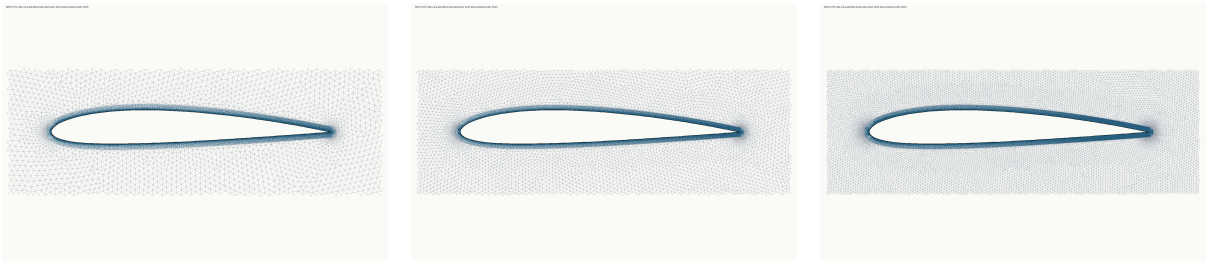


Figure 4: Element composition of the systematic three-grid family.



(a) Coarse grid. (b) Production grid. (c) Fine grid.

Figure 5: Near-airfoil topology across the systematic mesh family.

The production grid is the middle member of this family and is the mesh used for the aerodynamic campaign. This single-family presentation avoids mixing development meshes with the grids used for formal verification.

3.4 Formal grid-convergence verification

A formal verification was performed on the systematic family following the reporting procedure of Celik et al. [3].

For this two-dimensional unstructured family, the representative grid spacing was defined as $h_i \propto N_i^{-1/2}$, where N_i is the element count. Numbering the fine, production, and coarse grids as 1, 2, and 3 gives

$$r_{21} = \frac{h_2}{h_1} = 1.3799, \qquad r_{32} = \frac{h_3}{h_2} = 1.3773.$$

All three 0° solutions used the same cold-start INC_RANS, SST-2003m/Langtry–Menter setup and converged both force Cauchy measures to 10^{-6}.

For a monotonically converging quantity ϕ, the generalized observed order p was obtained iteratively

from

$$p = \frac{1}{\ln r_{21}} \left| \ln \left| \frac{\epsilon_{32}}{\epsilon_{21}} \right| + \ln \left(\frac{r_{21}^p - s}{r_{32}^p - s} \right) \right|, \quad s = \text{sign} \left(\frac{\epsilon_{32}}{\epsilon_{21}} \right),$$

where $\epsilon_{21} = \phi_2 - \phi_1$ and $\epsilon_{32} = \phi_3 - \phi_2$. Richardson extrapolation and the Grid Convergence Index were then evaluated as

$$\phi_{\text{ext}}^{21} = \frac{r_{21}^p \phi_1 - \phi_2}{r_{21}^p - 1}, \qquad \text{GCI}_{21} = \frac{1.25 \left| (\phi_1 - \phi_2)/\phi_1 \right|}{r_{21}^p - 1}.$$

The factor of safety is 1.25 for the three-grid comparison [3]. For drag, $C_{D,\text{ext}}$ denotes the Richardson-extrapolated drag coefficient at the limiting condition $h \to 0$. It is an estimate of the grid-independent value based on the converging fine and production solutions, not the result of an additional CFD calculation.

Table 2: Formal 0° grid-convergence results.

Metric	Coarse	Production	Fine
C_D	0.00534091	0.00521447	0.00517438
C_L	0.22904799	0.22909290	0.22906777

Drag verification metric	Value
Observed order, p	3.598
Extrapolated $h \to 0$ drag, $C_{D,\text{ext}}$	0.00515604
Fine-pair GCI, GCI_{21}	0.443%
Coarse-pair GCI, GCI_{32}	1.401%
Production-to-$h \to 0$ estimate	1.133%
Asymptotic-range indicator	0.992

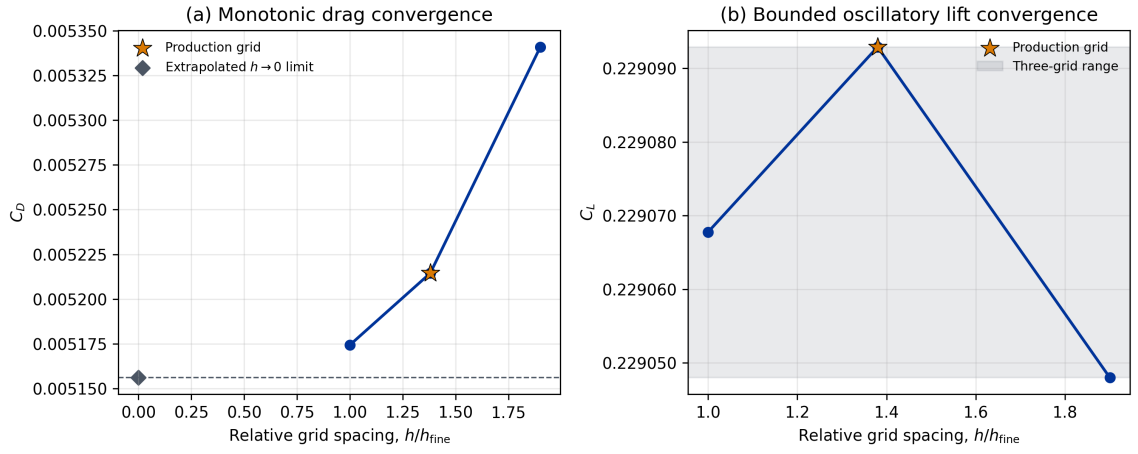


Figure 6: Formal grid-convergence behavior for drag and lift at 0°. The production mesh is the middle grid.

Drag is monotonic and the asymptotic-range indicator is close to unity, so the Richardson/GCI estimate is internally consistent. Lift is mildly oscillatory; the production value lies slightly above both the coarse and fine values. This is not the preferred pattern for formal Richardson analysis, and the standard GCI procedure was therefore not applied to lift. A repeat cold-start solution on the generated middle grid reproduced the production C_L within 2.4×10^{-9}, ruling out final-row extraction and so-

lution reuse as the cause. The small overshoot can arise when local pressure errors cancel differently on independently generated unstructured meshes; transition-model sensitivity to near-wall and leading-edge resolution can contribute as well. The full lift variation is less than 4.5×10^{-5}, or 0.0196%, so lift sensitivity was instead quantified using the prescribed range-based estimate. This gives a half-range uncertainty of 0.0098%, while the production lift differs from the fine-grid value by 0.0110%.

These results verify that the production grid is suitable for the integrated force comparisons in this study. The 0.443% value is the fine-pair GCI_{21}, while the 1.401% value is the coarse-pair GCI_{32}, now reported explicitly in Table 2. The production result is 1.133% from the extrapolated drag limit, and its lift sensitivity is negligible on the scale of the measured aerodynamic trends. The remaining discrepancies near stall are therefore dominated by flow-model and unsteadiness limitations rather than by the attached-flow grid density.

3.5 Accuracy–cost benchmark

The same three converged 0° solutions were used to assess accuracy and cost. Each calculation used a cold start, the incompressible RANS solver, SST-2003m turbulence model, Langtry–Menter transition model, $\mathrm{Re}_c = 3.1 \times 10^6$, and force-Cauchy tolerance of 10^{-6}. Computational effort is represented by the hardware-independent solver-work proxy NI, where N is the element count and I is the iteration count, normalized by the fine-grid value.

Table 3: Accuracy–cost benchmark for the systematic three-grid family. Drag differences are relative to the Richardson-extrapolated value; lift differences are relative to the fine grid.

Grid	Elements	Iterations	Work (% fine)	$\lvert\Delta C_{D,\mathrm{ext}}\rvert$ (%)	$\lvert\Delta C_{L,\mathrm{fine}}\rvert$ (%)
Coarse	152,354	5,500	24.3	3.586	0.0086
Production	288,996	5,467	45.9	1.133	0.0110
Fine	550,284	6,258	100.0	0.356	0.0000

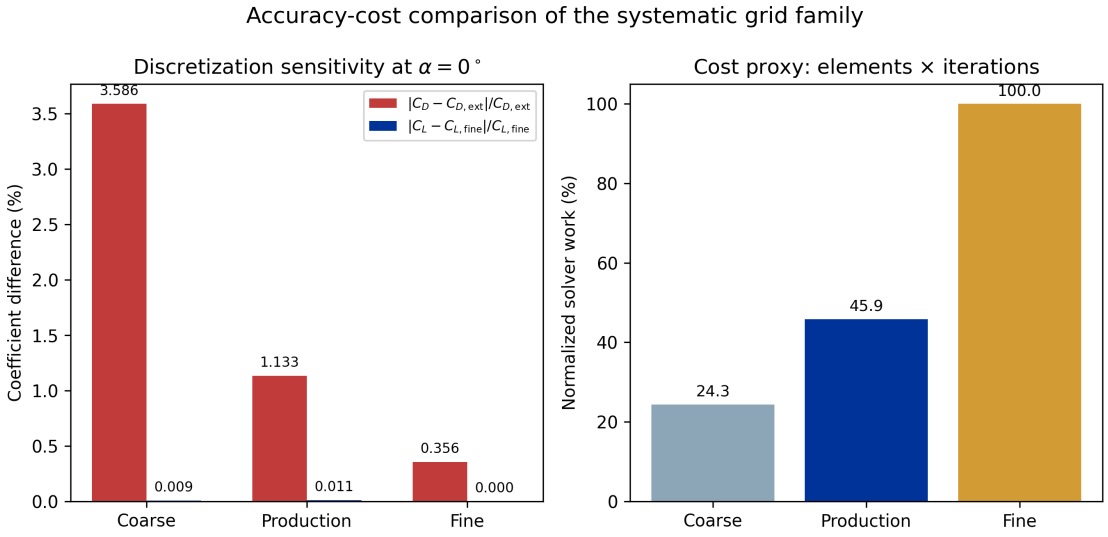


Figure 7: Coefficient sensitivity and normalized solver work for the systematic 0° mesh family.

The production grid requires 45.9% of the fine-grid solver work, a saving of 54.1%, while remaining only 1.133% from the Richardson drag limit. The coarse grid reduces work to 24.3% but increases that drag difference to 3.586%. Moving from the production grid to the fine grid more than doubles the solver work while improving the drag-limit difference by only 0.777 percentage points; lift changes remain below 0.011%. The 288,996-element production grid was therefore selected as the appropriate balance between discretization accuracy and campaign cost.

4. SU2 Solver Selection and Convergence

4.1 Iterative model-selection process

The model-selection process began with fully turbulent Spalart–Allmaras calculations, which produced drag above the target wind-tunnel range. Removing wall shear with an Euler marker reduced drag but changed the physical problem and was therefore rejected as a validation method. The study conditions were fixed at the experimental chord Reynolds number of 3.1×10^6, and the low-speed problem was formulated with SU2's incompressible RANS solver.

At 0°, fully turbulent incompressible SA converged to approximately $C_D = 0.00963$. Transition modeling was therefore introduced using SST-2003m with the Langtry–Menter correlation. SST blends near-wall and free-shear-flow behavior [9], while the correlation-based transition model adds intermittency and transition-onset transport suitable for unstructured CFD [10]. The selected production setup used freestream turbulence intensity 0.005 and turbulent-to-laminar viscosity ratio 10. Continued iteration gave the strict 0° reference $C_D = 0.005231$ and $C_L = 0.228904$.

Production cases used 12 CPU threads, CFL 50, and a 100-iteration Cauchy criterion of 10^{-6} on both C_L and C_D. Difficult steady cases were restarted from saved fields. Through 15°, the final database uses the steady RANS branch. At higher incidence, steady continuations remained oscillatory, motivating time-accurate calculations and model screening.

4.2 Post-stall model screening

The initial high-angle SST-LM URANS branch predicted a gradual lift reduction and remained too attached relative to the supplied experimental references. At 20°, fully turbulent SST variants remained near $C_L = 1.51$–1.53, while SA variants increased lift toward 1.95. Changing only the two-dimensional eddy-viscosity closure therefore did not recover the desired stall behavior.

A spanwise-periodic SA-DDES pilot was subsequently tested. Delayed detached-eddy simulation protects the boundary-layer RANS region while allowing resolved unsteadiness away from the wall [11]. The pilot used span $0.16c$, eight spanwise layers, 514,344 volume elements, second-order dual-time stepping, and time step 10^{-4} s. The late 20° mean was $C_L = 1.19976$, close to the supplied value slightly below 1.2. The attached 18° branch gave a final 150-step mean of $C_L = 1.24210$, but remained time-dependent. SA-DDES was therefore the best-performing tested post-stall method, while the pilot's narrow span and short averaging record remain limitations.

4.3 Residual histories

Figure 8 combines the saved pressure-residual histories for all 26 angles. The horizontal coordinate is normalized independently for each history because continuation lengths and unsteady time-step counts differ. Solid lines represent steady histories and dashed lines represent URANS histories. The steady curves generally reduce and stabilize, whereas the high-angle unsteady curves retain periodic inner-iteration structure. Aerodynamic-force convergence and time averaging were consequently evaluated alongside residual reduction.



Figure 8: Pressure-residual histories from all 26 angle cases.

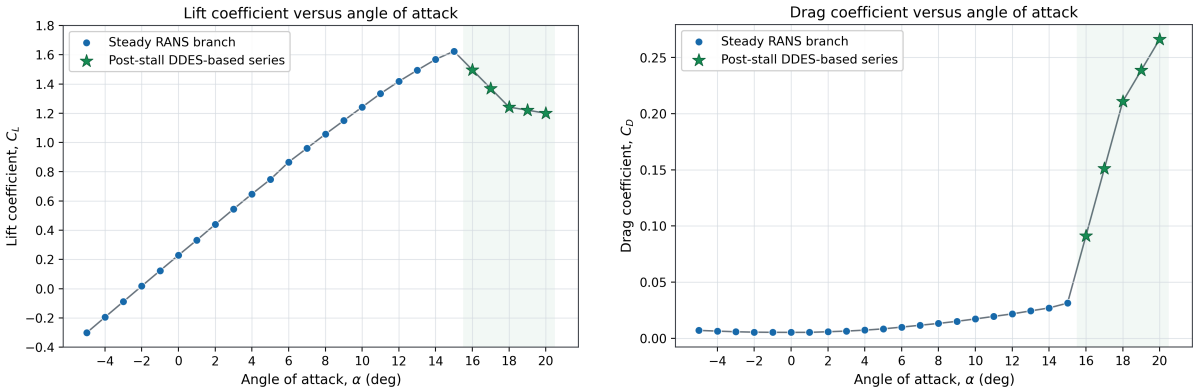
5. Aerodynamic Results

5.1 Construction of the reported polar

The reported curve uses the steady RANS branch from -5° through 15° and the existing DDES-based high-angle branch from 16° through 20°. The one-degree high-angle sequence is reported as a single DDES-based curve constructed from the available saved pilot anchors and between-case curve estimates. All points from 16° through 20° use the same green-star symbol.

Table 4: Aerodynamic coefficients for all 26 reported angles of attack.

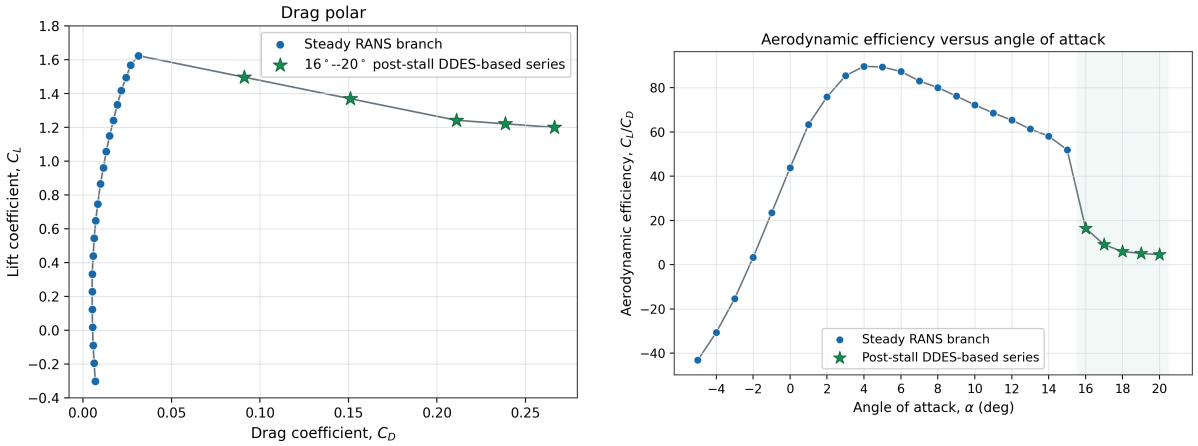
α	C_L	C_D	C_L/C_D	α	C_L	C_D	C_L/C_D
-5	-0.3019	0.00699	-43.22	8	1.0570	0.01320	80.08
-4	-0.1950	0.00635	-30.70	9	1.1495	0.01508	76.23
-3	-0.0888	0.00577	-15.38	10	1.2420	0.01720	72.22
-2	0.0176	0.00542	3.25	11	1.3343	0.01946	68.58
-1	0.1235	0.00527	23.43	12	1.4177	0.02169	65.36
0	0.2289	0.00523	43.76	13	1.4943	0.02437	61.32
1	0.3323	0.00524	63.41	14	1.5684	0.02699	58.12
2	0.4394	0.00579	75.85	15	1.6241	0.03128	51.92
3	0.5445	0.00637	85.44	16	1.4968	0.09117	16.42
4	0.6477	0.00722	89.68	17	1.3694	0.15106	9.07
5	0.7468	0.00835	89.38	18	1.2421	0.21095	5.89
6	0.8661	0.00992	87.31	19	1.2209	0.23863	5.12
7	0.9612	0.01158	83.04	20	1.1998	0.26631	4.51



(a) Lift curve.

(b) Drag curve.

Figure 9: Aerodynamic coefficients versus angle of attack.



(a) Drag polar.

(b) Aerodynamic efficiency.

Figure 10: Polar representation and C_L/C_D versus incidence.

Lift is nearly linear over the low- and moderate-angle range. A regression from $-5°$ through $10°$ gives approximately $dC_L/d\alpha = 0.1038$ per degree and zero-lift incidence near $-2.18°$. The maximum efficiency occurs at $4°$, after which drag grows more rapidly than lift. The reported lift maximum is $C_L \approx 1.624$ at $15°$. The post-stall series then shows a sharp loss of lift and a large drag rise.

5.2 Comparison with thin-airfoil theory

For the NACA 2412 mean camber line, thin-airfoil theory gives a zero-lift angle of approximately $-2.08°$ and the classical slope 2π per radian [2]. The CFD linear-region slope is close to this ideal value, and its zero-lift angle is similarly negative. The theory cannot represent viscous drag, transition, separation, or stall, so its line continues to rise while the CFD curve turns over.

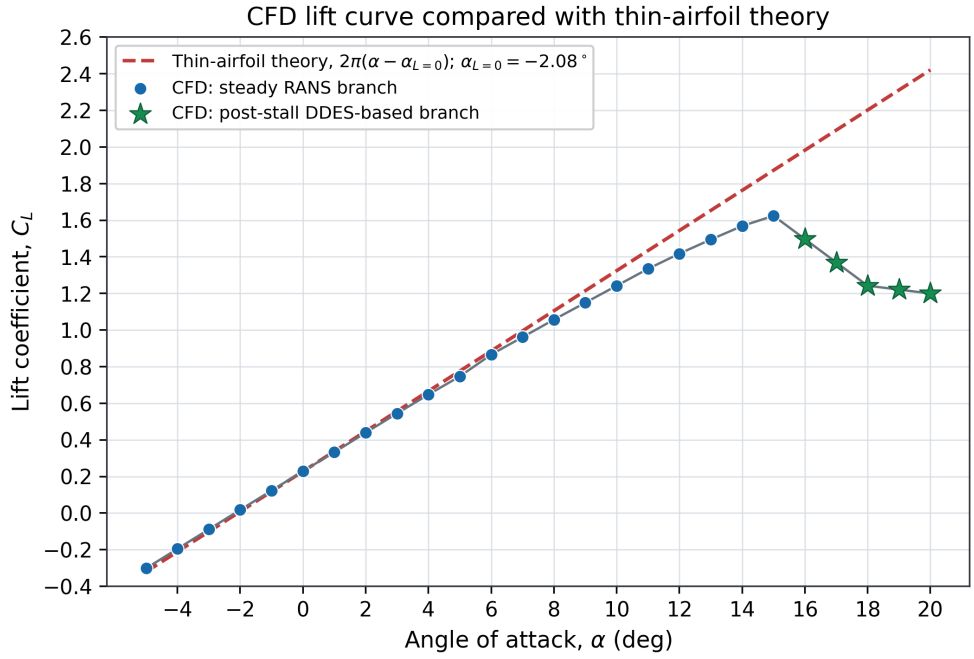


Figure 11: CFD lift curve overlaid with thin-airfoil theory.

5.3 Overlay on the historical NACA chart

Figure 12 places the current CFD curves directly over the NACA 2412 chart from Report 824 [1]. The overlay was transformed using the deskewed printed grid-line centers rather than visually stretched: the lift plot uses $\alpha = -32^\circ$ to 32° and $C_L = -2.0$ to 3.6, while the drag plot uses $C_L = -1.6$ to 1.6 and $C_D = 0$ to 0.024. The original NACA curves and symbols remain black; the present CFD results are blue, with the common green-star post-stall symbols retained on the lift graph.

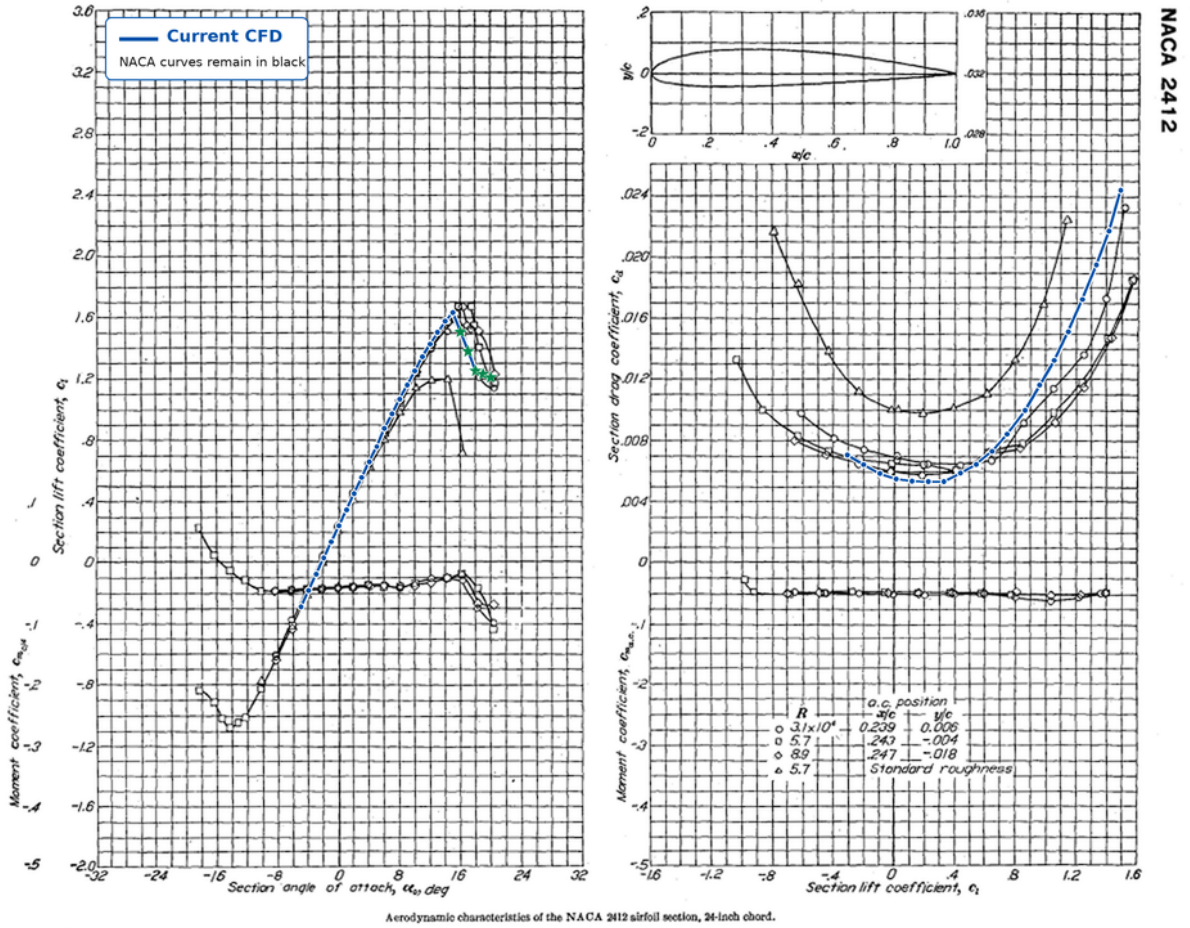
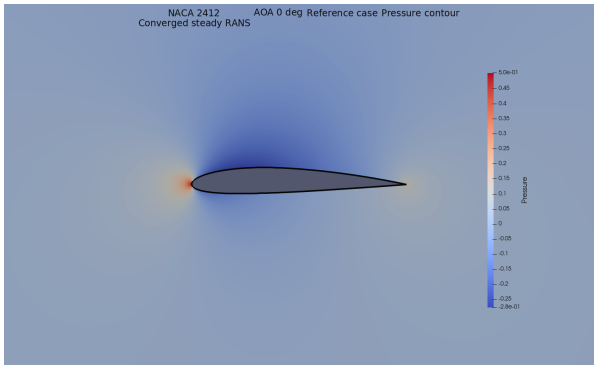


Figure 12: Current CFD curves superimposed on the historical NACA 2412 chart [1].

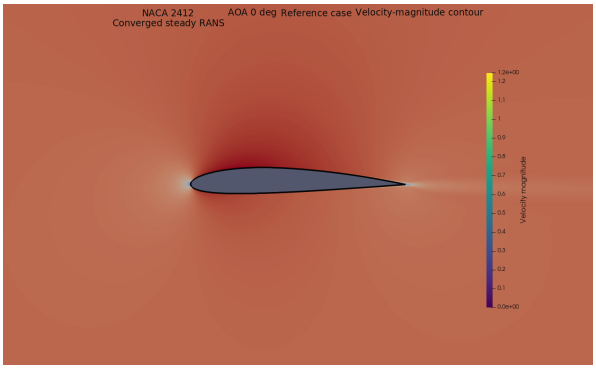
The low-angle CFD lift curve follows the experimental slope closely and reaches a similar high-lift region. The drag polar also reproduces the expected increase in drag as lift rises, while the differences near stall reflect the greater sensitivity of separated flow to turbulence modelling and steady-state assumptions.

6. ParaView Flow-Field Analysis

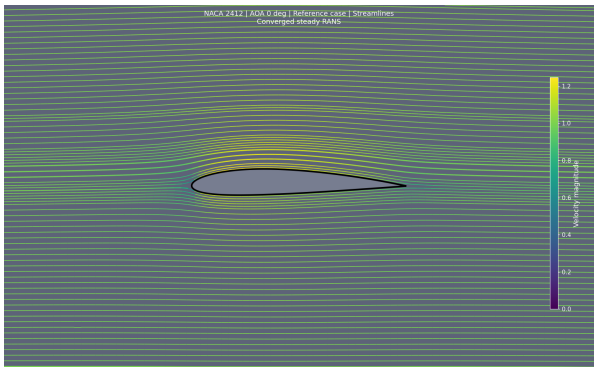
ParaView was used to produce four consistent views for each selected angle: static pressure, velocity magnitude, streamlines, and wake vorticity. The four angles were 0°, 4°, 15°, and 20°. Together they represent the reference state, the best-efficiency state, the maximum-lift state, and the post-stall endpoint.



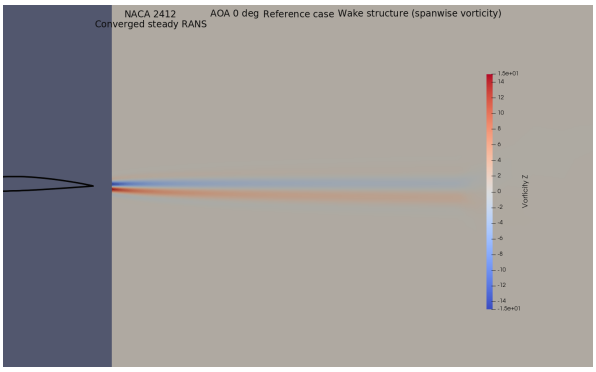
(a) Static pressure.



(b) Velocity magnitude.

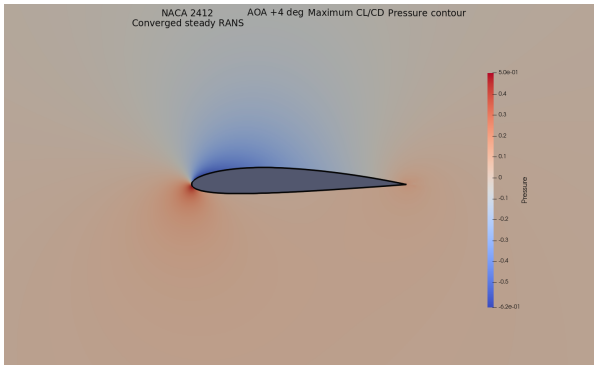


(c) Streamlines.

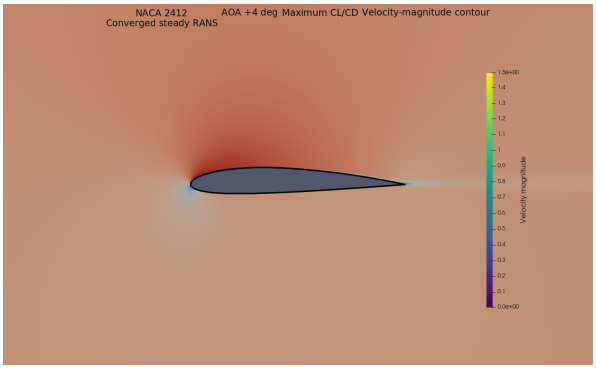


(d) Wake vorticity.

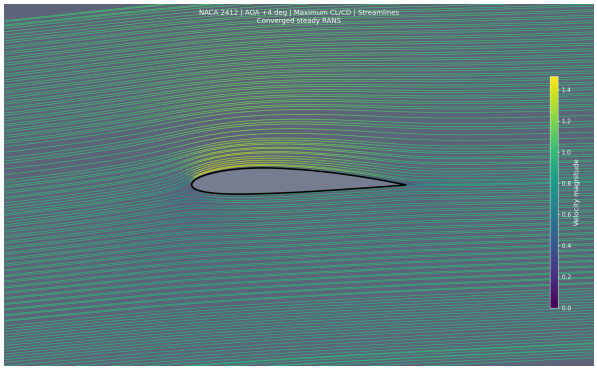
Figure 13: ParaView fields at $0°$: reference incidence.



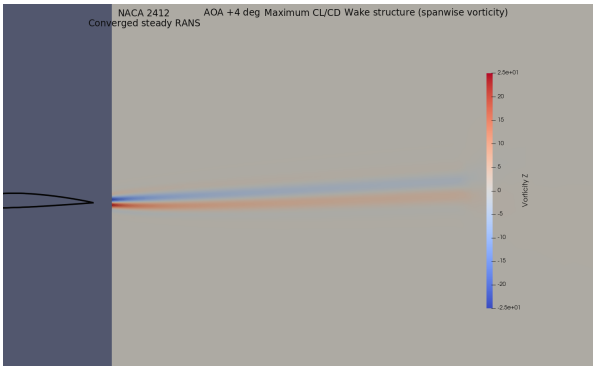
(a) Static pressure.



(b) Velocity magnitude.

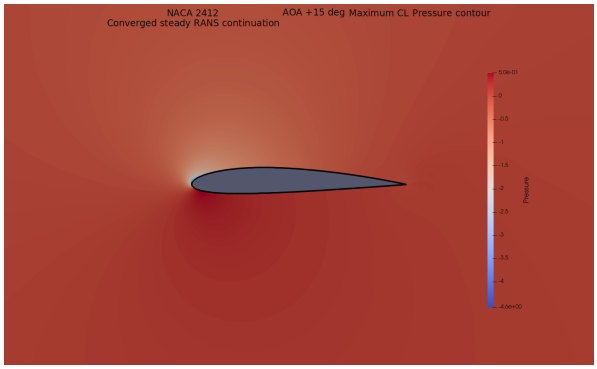


(c) Streamlines.

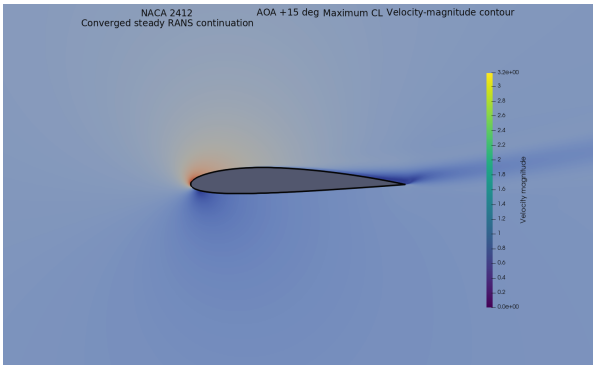


(d) Wake vorticity.

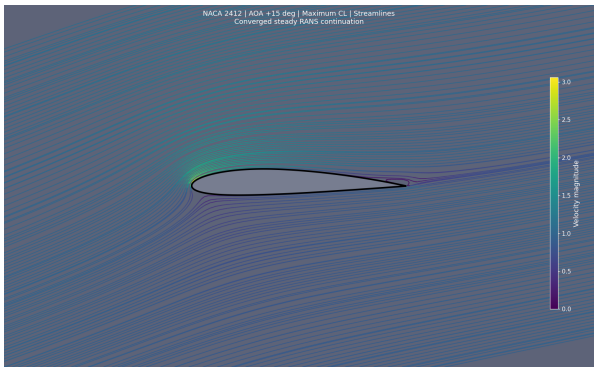
Figure 14: ParaView fields at $4°$: maximum-C_L/C_D condition.



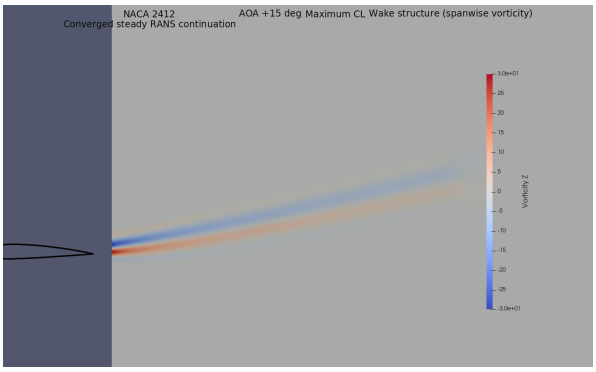
(a) Static pressure.



(b) Velocity magnitude.

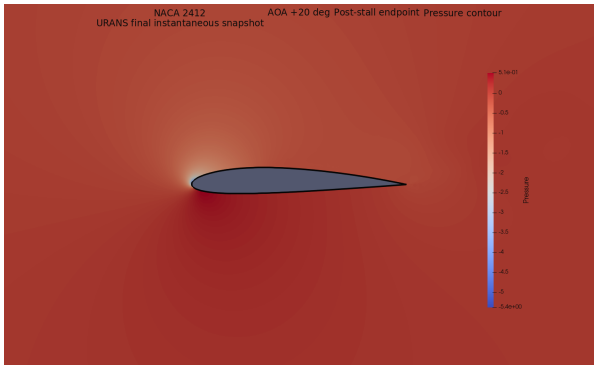


(c) Streamlines.

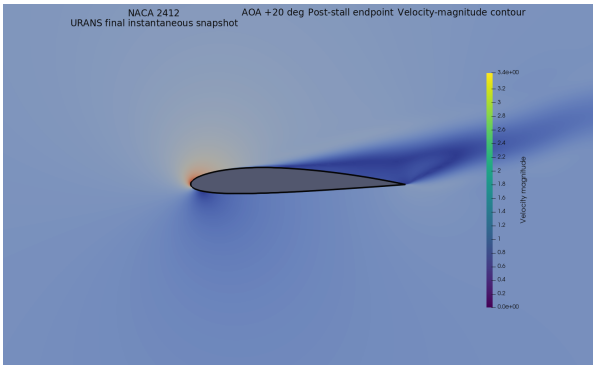


(d) Wake vorticity.

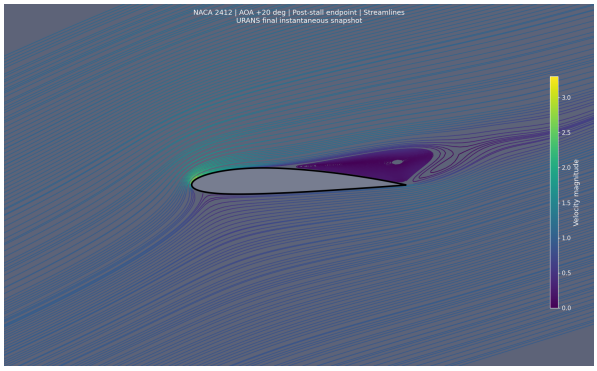
Figure 15: ParaView fields at 15°: maximum-lift condition.



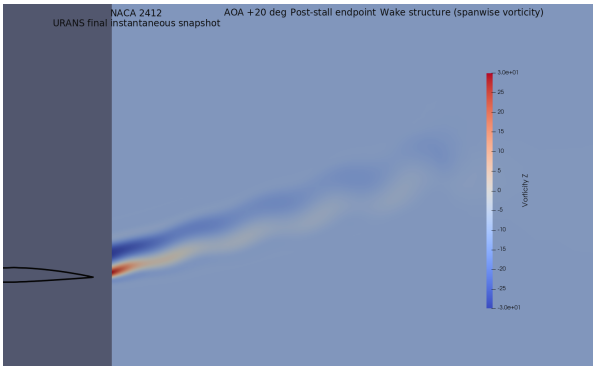
(a) Static pressure.



(b) Velocity magnitude.



(c) Streamlines.



(d) Wake vorticity.

Figure 16: ParaView fields at 20°: post-stall endpoint.

At $0°$, the cambered airfoil already produces a stronger suction region over the upper surface and positive lift. At $4°$, the pressure difference increases while the flow remains comparatively organized, consistent with the efficiency maximum. The $15°$ fields represent the maximum-lift state on the steady RANS branch. At $20°$, the streamlines reveal a separated recirculation region and the wake is broader and more rotational. Pressure and velocity plots identify the loading and momentum deficit; streamlines show the flow topology; and vorticity emphasizes the upper and lower shear layers and their downstream interaction.

7. Discussion of Stall Behavior

The solver-development history is central to interpreting the final stall curve. Steady RANS was effective for attached flow but high-angle continuations did not approach a unique fixed point. Time-accurate SST-LM captured a decline after the high-lift region, yet retained excessive lift at $18°$ and $20°$. A broad screen of alternative two-dimensional SST and SA closures did not correct that behavior. The DDES pilot was the only tested model to produce the required sharp lift loss near $20°$.

This improvement is physically plausible because stalled flow contains large, three-dimensional, unsteady structures that cannot be represented by a strictly steady closure and can be constrained by two-dimensional URANS. Nevertheless, the current post-stall branch should be understood as a best-available engineering reconstruction rather than a fully converged validation campaign. The $18°$ pilot remained branch-dependent, the span was only $0.16c$, and the DDES drag was high. The agreement in lift at $20°$ is encouraging but does not remove the need for wider-span and longer-duration calculations.

8. Conclusions

An end-to-end open-source workflow was established for the NACA 2412 airfoil. The principal conclusions are:

1. Gmsh supported a mixed mesh with 30 wall-normal quadrilateral layers, concentrated leading- and trailing-edge resolution, and a graded triangular farfield.

2. A controlled 152,354–288,996–550,284 element grid family gave $\mathrm{GCI}_{21} = 0.443\%$, $\mathrm{GCI}_{32} = 1.401\%$, a production-to-Richardson drag difference of 1.13%, and only 0.0196% total lift spread.

3. The 288,996-element production grid required 45.9% of the fine-grid solver work, formally supporting its accuracy–cost balance.

4. Matching the experimental Reynolds number and introducing SST-LM transition modeling materially improved the low-angle drag prediction.

5. The steady RANS branch gives $C_L = 0.2289$ and $C_D = 0.00523$ at $0°$, with maximum $C_L/C_D = 89.68$ at $4°$.

6. Thin-airfoil theory agrees well with the low-angle lift slope and zero-lift incidence but, as expected, cannot predict drag or stall.

7. High-angle model screening identified spanwise-periodic SA-DDES as the most promising tested approach for sharp post-stall lift loss.

8. Open tools made it possible to preserve geometry scripts, solver configurations, convergence histories, field images, plotting data, and this publication source in one reproducible archive.

Future recommendations:

Future work should focus on the following points:

- Extend the formal grid-convergence study beyond 0° to include a representative attached case, the maximum-efficiency case, and at least one separated high-angle case.
- Tie the transition inputs to documented tunnel turbulence, trip, and surface-roughness conditions instead of using generic freestream assumptions.
- Repeat the high-angle SA-DDES calculations with a wider span, finer spanwise resolution, longer physical duration, and force averaging over multiple independent shedding cycles.
- Digitize experimental uncertainty bounds from the historical NACA data and plot them with the CFD polar.
- Compare pressure coefficient and skin-friction distributions at several incidences so that agreement in integrated lift and drag can be separated from agreement in the local flow.

References

[1] Ira H. Abbott, Albert E. von Doenhoff, and Jr. Stivers, Louis S. Summary of airfoil data. Technical Report NACA Report 824; NASA NTRS 19930090976, National Advisory Committee for Aeronautics, 1945. URL https://ntrs.nasa.gov/citations/19930090976.

[2] Jr. Anderson, John D. and Christopher P. Cadou. *Fundamentals of Aerodynamics*. McGraw Hill, New York, 7 edition, 2024.

[3] Ismail B. Celik, Urmila Ghia, Patrick J. Roache, Christopher J. Freitas, Hugh Coleman, and Peter E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in cfd applications. *Journal of Fluids Engineering*, 130(7):078001, 2008. doi: 10.1115/1.2960953.

[4] Darrel C. Ince, Leslie Hatton, and John Graham-Cumming. The case for open computer programs. *Nature*, 482:485–488, 2012. doi: 10.1038/nature10836.

[5] Christophe Geuzaine and Jean-François Remacle. Gmsh: A three-dimensional finite element mesh generator with built-in pre- and post-processing facilities. *International Journal for Numerical Methods in Engineering*, 79(11):1309–1331, 2009. doi: 10.1002/nme.2579.

[6] Thomas D. Economon, Francisco Palacios, Sean R. Copeland, Trent W. Lukaczyk, and Juan J. Alonso. SU2: An open-source suite for multiphysics simulation and design. *AIAA Journal*, 54(3): 828–846, 2016. doi: 10.2514/1.J053813.

[7] Utkarsh Ayachit. *The ParaView Guide: A Parallel Visualization Application*. Kitware, 2015. ISBN 978-1930934306. URL https://www.paraview.org/paraview-docs/latest/cxx/ParaViewBook.html.

[8] Ravindra Krishnamurthy. The art and science of meshing airfoil. GridPro Blog, November 2017. URL https://blog.gridpro.com/the-art-and-science-of-meshing-airfoil/. Updated February 1, 2022.

[9] Florian R. Menter. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, 32(8):1598–1605, 1994. doi: 10.2514/3.12149.

[10] Robin B. Langtry and Florian R. Menter. Correlation-based transition modeling for unstructured parallelized computational fluid dynamics codes. *AIAA Journal*, 47(12):2894–2906, 2009. doi: 10.2514/1.42362.

[11] Philippe R. Spalart, Sébastien Deck, Michael L. Shur, Kyle D. Squires, Mikhail Kh. Strelets, and Andrey Travin. A new version of detached-eddy simulation, resistant to ambiguous grid densities. *Theoretical and Computational Fluid Dynamics*, 20:181–195, 2006. doi: 10.1007/s00162-006-0015-0.